

MicroTouch™

- Aliphatic Polyurethane Beads
- Spherical Shape
- Range of particle sizes (F,VF,XF)
- Transparent - Colorless
- Highly Resilient (elongation)
- Virtually Indestructible
- No discoloring on aging/UV



Benefits of MicroTouch™ Beads

- **Imparts pleasing tactile properties**
 - Natural look and feel to finishes
 - Soft to the touch and matted - silky smooth
- **Helps low cost surfaces feel luxurious**
 - Plastics, wood, paper and glass
 - Glossy can be associated with cheap

Side Benefits of PU beads

- **Burnish/Mar Resistance**
 - Can self heal (elongation)
 - Very important on flexible substrates
 - Will not detach from the film
- **Gloss Reduction w/excellent film clarity**
 - Not as efficient as PropylMatte or MicroMatte
 - More expensive then polypropylene matting wax

PRODUCT GUIDE

Typical Properties	MicroTouch 800F	MicroTouch 800VF	MicroTouch 800XF	MicroTouch 850XF
Density @ 77 °F (25 °C)	1.05	1.05	1.05	1.02
Maximum Particle Size (µm)	120	60	31	31
Mean Particle Size (d50, µm)	22-30	11-15	6-9	5-9

The MicroTouch™ 850 Series

- MicroTouch 850XF
 - Synergistic combination of polyurethane beads and high density polyethylene wax (HDPE)
 - Mean particle size 4–8 μm
 - Maximum particle size 31 μm
- More economical than 800 series
- Better scratch resistance
- Silkier surface feel

Market Success Stories

- 800XF – WB gravure ink for Frito Lay packaging
- 800XF/850XF – Interior automotive – SB urethanes
- 800XF – WB urethane coating for shoe soles

Most Commercial Success

- 100% UV Acrylate Varnishes For Graphic Arts



Formulation Considerations

- Resin selection is critical for soft feel
 - Low Tg – WB, SB, and UV.
- 10-20% loadings required for on Resin Solids
- Micromide 520 or Superglide 904 can be used for lubricity/silkiness
- When used with higher Tg resins, will impart smooth consistent matting, silky feel and and good burnish resistance.

Performance and Percent Addition

- **1-2% TFW** - Improve Burnish Resistance
- **2-6% TFW** - Gloss Reduction with Minimal Tactile Properties
- **6-10% TFW** - Best Tactile Effects and Burnish Resistance
- Best practice – High Speed Dispersion

Polar Solvents and Swelling

- Will swell if dispersed directly into alcohols
 - Ethanol and isopropanol
- Minimize the effect by dispersing the beads directly into the resin prior

Temperature Limitations

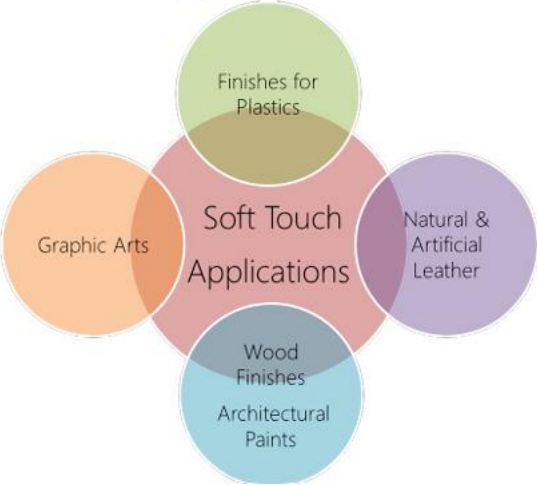
- Not recommended for baking applications
- Applications with curing temperatures of **100-120° C** are fully acceptable (10-20 minutes)
- Application between **120-140°C** - MicroTouch beads may dis-colour
 - thermal stabilizers can be added to alleviate this
- MicroTouch is not recommended for application temperatures **>140°C**

Sales/Marketing Aspects of Soft Touch

- Making Ordinary Products Stand Out



Soft Touch Applications



Plastics



Graphic Arts



Leather and Textiles



Wood & Architectural



Other Options for Soft Touch

- PropylTex Series – 325S through 270S
 - Highly branched imparts a unique “dry” feel
 - Good resistance to burnish and gloss up
- PropylTex HD series
 - Same performance
 - Better in-can stability

